8 Fig. 8.1 shows an electrical circuit.

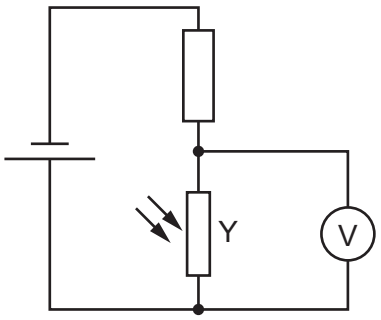


Fig. 8.1

(a) The light intensity at the circuit increases from dark to bright.

State any effect on the resistance of component Y.

.....

State and explain any effect on the reading of the voltmeter.

.....

.....

[3]

(b) The circuit shown in Fig. 8.2 is switched on for 2.0 min.

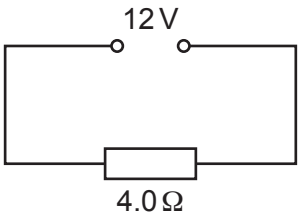


Fig. 8.2

The current in the 4.0 Ω resistor is 3.0 A and the magnitude of the charge on an electron is 1.6×10^{-19} C.

(i) Calculate the number of electrons that pass through the resistor each second.

number = ... [3]

(ii) Calculate the power dissipated by the resistor.

power = ... [2]

[Total: 8]

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